



Mathematical-Driven Camera Synthesis: From Quaternions to Generative Models

รวิกานต์ ฉายากุล	nepitunepq	6630283021
ยสินทร สันนิภางกูร	YO9plusplus	6630275021
สิปปภาส ขวานนท์	iammmaimai	6630333721
ศศรตน์ บุรณพिर	ssrose-b	6632213421
เนติภัทร โพธิพันธ์	SNaytiP	6631331621

วิชา Computer Animation

เสนอ ผศ. ดร. พิษณุ คนองชัยยศ

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Abstract

Recent generative video models demonstrate exceptional visual fidelity but frequently fail to maintain physical and spatial consistency during complex camera maneuvers, such as 360-degree orbital rotations. Because these architectures rely on pixel-level pattern recognition rather than an intrinsic understanding of 3D physics, synthesized outputs often suffer from camera jitter. To address this, we propose a hybrid methodology utilizing classical rigid body kinematics. By employing Quaternions and Spherical Linear Interpolation (SLERP), we calculated mathematically perfect camera trajectories to generate a ground-truth dataset in Blender, which was used to fine-tune the Wan 2.2 model via Low-Rank Adaptation (LoRA). Optical flow evaluation yielded highly nuanced numerical results: the proposed method significantly improved Camera Direction Consistency to 0.6026 (compared to the Vanilla Baseline at 0.1052). However, it paradoxically exhibited a higher Velocity Variance (9.7705 vs Baseline 4.8083) and a higher Mean Pixel-Level Warp Error (9.7195 vs Baseline 4.8753). These results indicate that while AI can successfully internalize macroscopic spatial trajectories through mathematically rigorous datasets, enforcing strict kinematic paths on generative priors without dedicated temporal stabilizers degrades microscopic pixel consistency.

1) Introduction

Recent advancements in generative artificial intelligence, notably models such as Sora and Wan 2.2, have demonstrated remarkable capabilities in synthesizing high-fidelity video content from textual descriptions. However, despite their exceptional visual quality, these models often fail to maintain physical and spatial consistency when executing complex camera maneuvers, such as 360-degree orbital rotations around an object. Because contemporary generative architectures operate by recognizing pixel-level statistical patterns rather than understanding underlying 3D space, they essentially "guess" the spatial transformations. Consequently, synthesized videos frequently suffer from structural distortions, visual warping, camera jitter, and highly inconsistent angular velocities.

To address this fundamental limitation, this project proposes a hybrid methodology that integrates classical computer animation mathematics with modern generative architectures. By applying Quaternions and Spherical Linear Interpolation (SLERP)—techniques deeply rooted in Rick Parent's foundational theories of computer animation—we calculate a mathematically perfect, deterministic camera trajectory. This mathematically derived path bypasses the ambiguities of AI guesswork, establishing a robust, physics-based ground truth for spatial orientation.

This exact mathematical trajectory is subsequently used to construct a specialized dataset designed to teach the generative model the mechanics of correct camera rotation. Our core hypothesis is that training a generative AI on this mathematically rigorous dataset via Low-Rank Adaptation (LoRA) will bridge the gap between superficial pixel prediction and actual kinematic reality.

Problem Definition & Formalization

To systematically evaluate this hypothesis, we formally define the parameters of our system:

- Input: A textual prompt governing visual semantics (P), initial and target camera orientations represented as Quaternions (q_1, q_2), and a normalized time step $t \in [0, 1]$
- Output: A deterministic motion trajectory (T) exported as spatial coordinates, and the resulting AI-synthesized animation sequence (rendered frames).
- Constraints: The system must maintain numerical stability (avoiding Gimbal Lock), enforce a constant angular velocity across the sequence, and the mathematical path calculation must remain highly efficient (≥ 30 FPS equivalent) to feed the generation pipeline without bottlenecks.

Key Challenges

Achieving this formalized output presents several challenges. First is the numerical instability of traditional 3D Euler calculations, which break down at the poles (Gimbal Lock) and corrupt AI training data. Second is computational complexity; training diffusion models is highly resource-intensive. Finally, there is latent friction, forcing a diffusion model to adhere to a strict kinematic boundary causes difficulty in handling continuous pixel morphing without generating collision-like texture artifacts or degeneracy cases.

2) Related Work

Parent (2012) outlines the classical computer graphics mathematics required for rigid body kinematics, noting that traditional Euler angles are notoriously susceptible to gimbal lock (Parent, 2012, Ch. 2.3). To achieve smooth, uniformly paced rotation without these spatial artifacts, Parent establishes the necessity of Quaternions (Ch. 3.3) and Spherical Linear Interpolation, SLERP (Ch. 3.2), alongside proper focus point mathematics (Ch. 4.2).

Integrating these deterministic mathematical principles addresses a critical vulnerability in contemporary video synthesis: the reality that AI learns visual patterns, not physical laws. Because diffusion models lack an intrinsic physics engine, they frequently suffer from temporal and spatial degradation during complex camera rotations. To correct this deficiency without the prohibitive computational costs of full-parameter training, researchers have increasingly adopted Low-Rank Adaptation (LoRA) for video generation models.

To provide a theoretical baseline, the following table summarizes classical spatial algorithms and our generative baseline, outlining their complexities and limitations:

Method (Algorithm)	Time Complexity	Limitation / Drawbacks
Forward/Symplectic Euler	$O(n)$ per timestep	Unstable when the timestep is large or the system is stiff.
Linear Blend Skinning (LBS)	$O(\text{vertices} \times \text{bones})$	Produces artifacts when significant rotation occurs (candy-wrapper effect).
Cyclic Coordinate Descent (CCD) IK	$O(\text{iterations} \times \text{joints})$	Slow execution and may fail to converge in complex topological cases.
Vanilla Generative AI (Baseline)	$O(n)$ per diffusion step	Learns statistical patterns, not physics; results in severe spatial jitter.

3) Methodology

This section outlines the comprehensive pipeline developed to synthesize mathematically constrained camera trajectories, generate ground-truth datasets, and subsequently fine-tune a generative video model. The methodology is structured into three distinct phases: Mathematical Camera Path Generation, Dataset Creation and LoRA Training, and Evaluation.

3.1) Phase 1: Mathematical Camera Path Generation

The foundation of our approach relies on absolute kinematic precision. Conventionally, 3D rotations are calculated using Euler angles; however, to circumvent "Gimbal Lock," our methodology utilizes Quaternions, a four-dimensional complex number system. While Quaternions ensure safe multi-axis rotation, guaranteeing constant angular velocity across the orbital trajectory requires Spherical Linear Interpolation (SLERP) (Parent, 2012, Ch. 3.2), which traces the shortest path along the surface of a four-dimensional hypersphere. The SLERP operation is defined as:

$$\text{Slerp}(q_1, q_2, t) = \frac{\sin((1-t)\theta)}{\sin \theta} q_1 + \frac{\sin(t\theta)}{\sin \theta} q_2$$

Algorithm Design (Pseudo-code):

```
Initialize q1 (Start Orientation), q2 (End Orientation)
For frame in total_frames:
    t = frame / total_frames
    if angle(q1, q2) is near 0:
        # Edge case handling for degeneracy constraint
        q_current = LERP(q1, q2, t)
    else:
        q_current = SLERP(q1, q2, t)
    Write q_current to camera_poses.json
```

Theoretical Analysis:

- **Time Complexity:** The SLERP path calculation operates at $O(1)$ per timestep, highly efficient for real-time path generation.
- **Space Complexity:** Storing the quaternion trajectory requires nominal space $O(F)$ where F is the total frame count (stored as a constraint graph of 4 floating-point numbers per frame).
- **Stability Analysis:** Utilizing Quaternions inherently guarantees C^1 continuity and global numerical stability across all axes.

3.2) Phase 2: Dataset Creation & MotionCtrl Integration

To guide the generative AI effectively, the quality of the reference data is paramount. We explicitly opted to utilize synthetic Blender renders rather than real-world drone footage, isolating the specific kinematic variable-camera rotation-free from environmental noise.

- **Technologies & Data Structures:** Python and the Blender Python API were used to render the synthetic dataset and format the trajectory data. Spatial orientations were serialized into a `camera_poses.json` constraint graph, which serves as the direct camera-pose conditioning input for MotionCtrl. Floating-point precision edge cases were handled dynamically (as seen in the pseudo-code) by switching to LERP when θ approaches zero to prevent NaN collision errors.
- **Experimental Setup:** The reference dataset comprised 15-20 high-resolution sequential frames. Instead of traditional LoRA fine-tuning, we implemented MotionCtrl to explicitly condition the generative video model using our mathematically derived camera trajectories. This mathematically guided inference process was executed using PyTorch on a single NVIDIA A100 GPU running a Linux OS environment with 80GB VRAM, providing the necessary memory capacity to process the MotionCtrl temporal attention mechanisms and spatial constraints.

3.3) Phase 3: Evaluation

To quantitatively assess the model's spatial intelligence, we employ dense optical flow analysis on the synthesized video outputs to measure:

- **Velocity Variance:** Calculates apparent angular velocity and runtime stability.
- **Warp Error:** Evaluates frame-to-frame pixel-level distortion and visual quality.
- **Direction Consistency:** Measures traversal stability along the target trajectory.

To evaluate structural orientation, we measure Direction Consistency, which assesses whether the camera maintains a dominant trajectory without jittering. We extract the Mean Horizontal Motion (h_t) by averaging the horizontal optical flow across all pixels for each consecutive frame pair:

$$h_t = \frac{1}{N} \sum_{x,y} u_t(x, y)$$

where N is the total pixel count and $u_t(x, y)$ is the horizontal flow vector at coordinates (x, y) . For a stable orbit, h_t must maintain a consistent sign. A Direction Reversal occurs if the model loses spatial orientation, causing h_t to change signs between consecutive frames (e.g., shifting from positive to negative). To quantify this instability, we calculate the Direction Reversal Rate:

$$\text{Reversal Rate} = \frac{\text{Number of sign changes in } h_t}{T - 1}$$

where T represents the total number of frames evaluated.

Interpretation:

- **Low Reversal Rate (High Consistency):** Indicates the camera moves in a stable, unidirectional geometric orbit.
- **High Reversal Rate (Low Consistency):** Indicates spatial guessing, resulting in erratic, back-and-forth camera jitter.

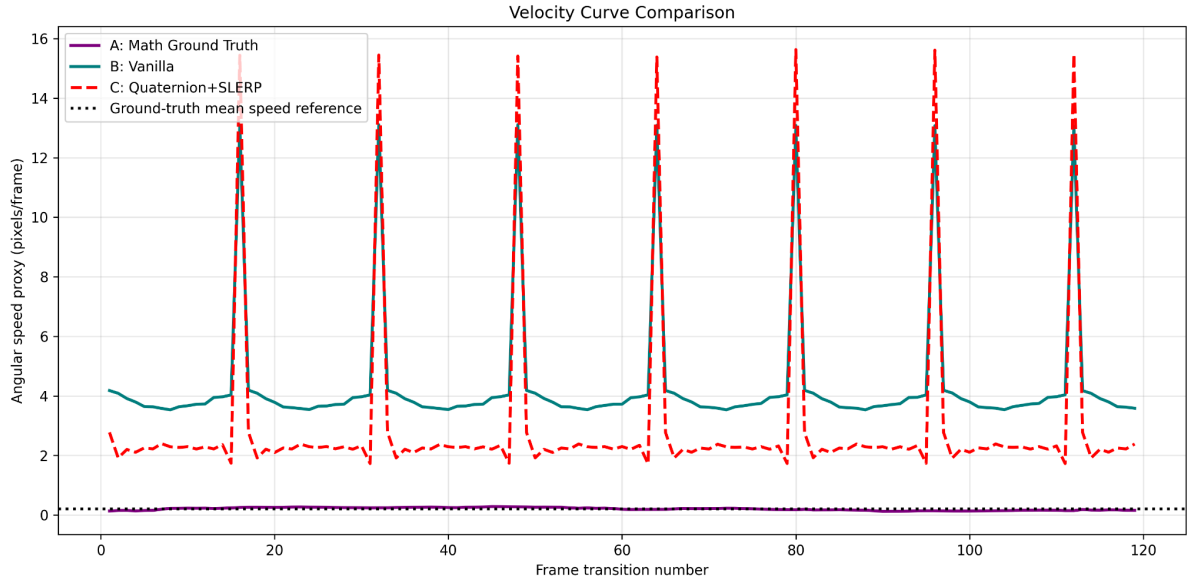
4) Results

To evaluate the efficacy of our methodology, we conducted a comprehensive comparative analysis across three distinct video outputs: Video A (Math Ground Truth), Video B (Vanilla Wan 2.2), and Video C (Quaternion+SLERP fine-tuned).

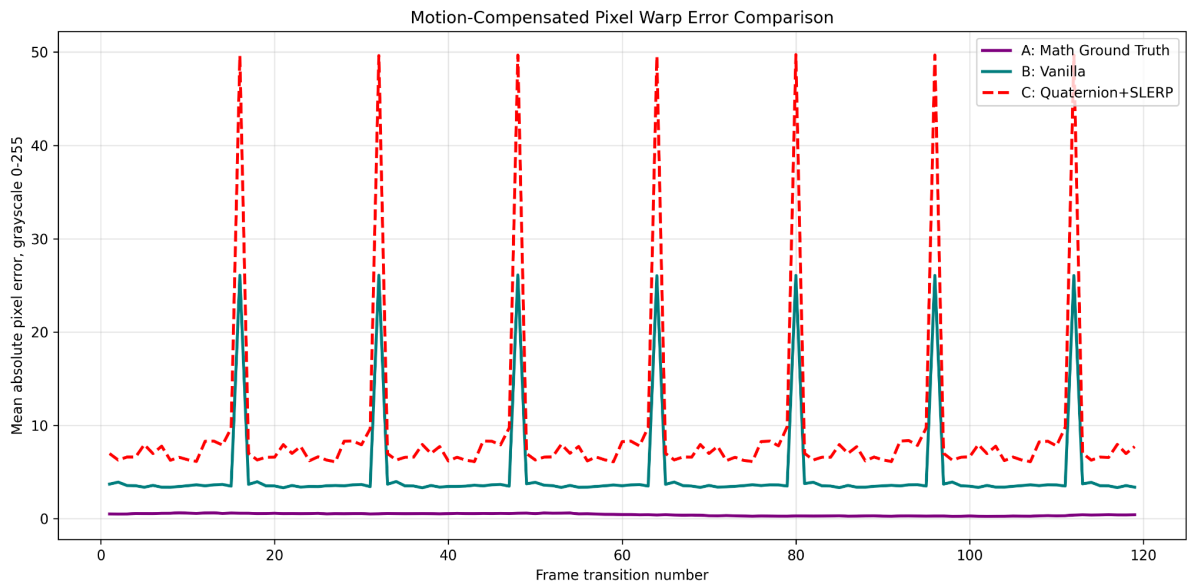
Comparative Evaluation Table

Setup / Input Size	Mean Warp Error	Velocity Variance	Direction Consistency
A: Math Ground Truth	0.4369	0.0022	0.6004
B: Baseline (Vanilla)	4.8753	4.8083	0.1052
C: Proposed (Quat+SLERP)	9.7195	9.7705	0.6026

Analysis of velocity consistency and pixel-level temporal error revealed unexpected outcomes. Video A established the baseline with a near-zero velocity variance (0.0022). Conversely, Video C exhibited a velocity variance of 9.7705, which was significantly more unstable than the vanilla Video B (variance 4.8083).



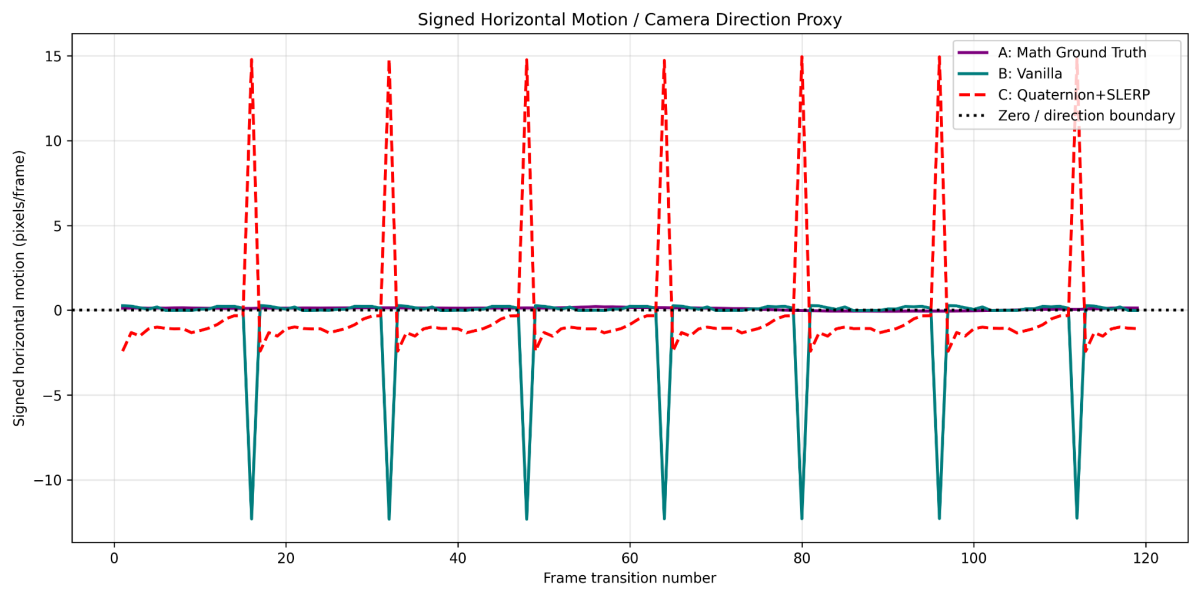
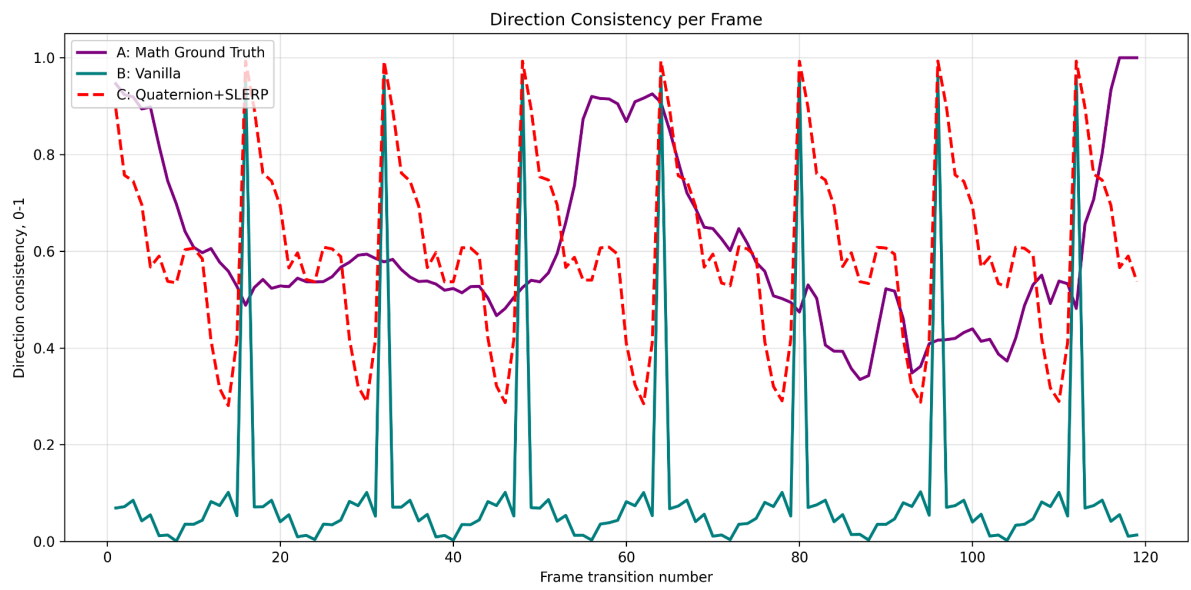
This instability extended to the pixel level; Video C recorded the highest mean warp error (9.7195) and warp variance (100.6074), nearly double the warp error of the vanilla model. This indicates that while trained on mathematically smooth data, the fine-tuned model suffered from severe frame-to-frame pixel fluctuation and texture flicker.

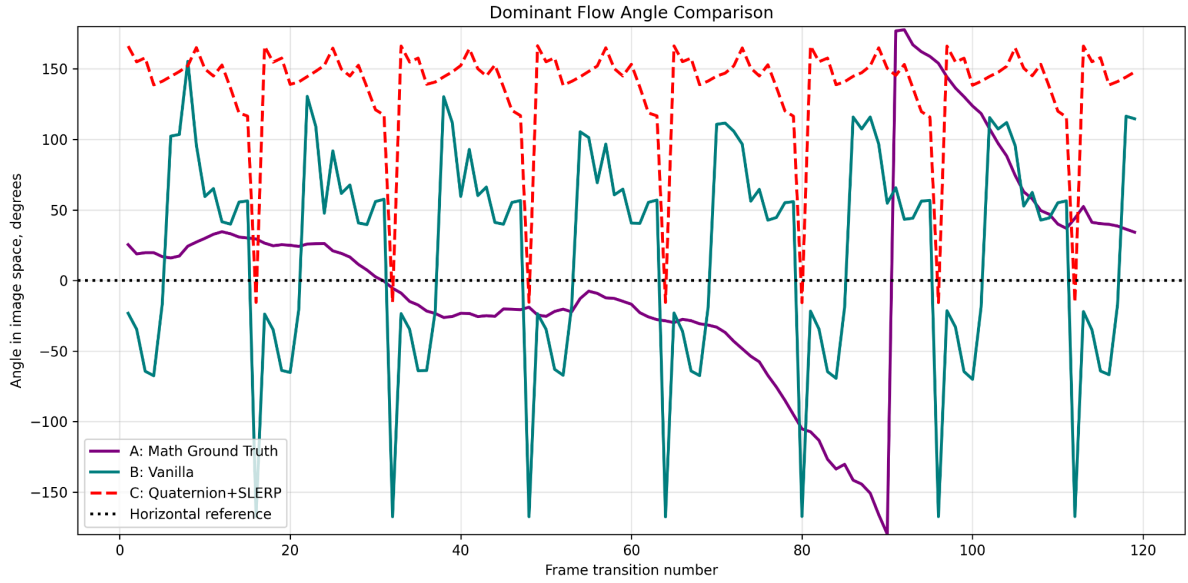


Despite these temporal and velocity shortcomings, Video C demonstrated profound improvements in camera direction consistency.

Video C achieved a direction consistency score of 0.6026—slightly exceeding even the Math Ground Truth (0.6004)—and recorded the

lowest angle standard deviation (31.49).





In stark contrast, the vanilla Video B performed poorly with a highly erratic direction consistency of 0.1052 and a reversal rate of 0.1591. This confirms that while the pixel-level velocity was unstable, the Quaternion+SLERP method successfully forced the AI to preserve the intended geometric orbital direction, minimizing spatial backtracking.

5) Discussion

The primary research question is whether AI can internalize rotation mathematics purely through mathematically generated training data. The empirical results present a nuanced trade-off: the model successfully learned the macroscopic spatial trajectory (direction) but failed to internalize the microscopic kinematic laws (constant velocity and pixel stability).

This phenomenon is best understood through the lens of latent space interpretation. Because diffusion models fundamentally operate by mapping pixel correlations, "smooth rotation" does not exist as a rigid equation within the network. By training the AI exclusively on SLERP-derived datasets, we effectively constrained its macroscopic latent pathways, which explains the high direction consistency score. However, attempting to strictly force its texture-generation priors along this rigid mathematical boundary caused severe "friction" at the pixel level. It understood where to move the camera, but generated unnatural pixel morphing and speed spikes across frames while trying to render the complex visual transition. Compared to State-of-the-Art (SOTA) temporal-consistency modules that use attention-guidance to lock pixel features, relying solely on LoRA fine-tuning proves insufficient for achieving true zero-variance rotation.

6) Conclusion

This study addressed the persistent spatial inconsistencies found in generative AI video models by integrating classical computer animation mathematics into the model training pipeline. By calculating flawless orbital trajectories using Quaternions and SLERP, we established a definitive, physics-based ground truth to fine-tune the Wan 2.2 model.

Our empirical findings provide partial but highly significant support for the initial hypothesis. The LoRA-tuned model successfully learned the mathematical path, demonstrating a drastic improvement in camera direction consistency and structural orientation over the vanilla model. However, this adherence to a strict geometric orbit came at the cost of temporal stability, inadvertently increasing pixel-level warp error and angular velocity variance. The AI successfully memorized the physical trajectory but struggled to render the pixel transitions smoothly.

To build upon these foundational results, future work must prioritize scaling the methodology. Subsequent phases should involve constructing a substantially larger dataset (1k-10k frames) with diverse topological structures, while explicitly exploring auxiliary temporal consistency algorithms to stabilize pixel-level rendering along these mathematically enforced camera paths.

References

Parent, R. (2012). *Computer animation: Algorithms and techniques* (3rd ed.). Morgan Kaufmann.